

Program: Diploma in Integrated Circuits Design & Fabrication	
Course Code: 6379A	Course Title: Industrial Automation & PLC Lab
Semester: 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on training to construct electric power converters and controllers using power semiconductor switching devices.
- To develop PLC ladder programs to realize real time applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Experiment with electronic components	2039	Fundamentals of Electrical & Electronics Engineering Lab	2
Electric circuits and network theorems	3041	Electric Circuits & Networks	3
Amplifiers and oscillators	3047	Electronic Circuits Lab	3
Logic circuits	3048	Digital Electronics Lab	3

Course Outcomes:

On completion of the course, the students will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Construct various gate triggering methods for SCR	12	Applying
CO2	Construct controlled rectifiers and power controlling circuits using power semiconductor switching devices	9	Applying
CO3	Implement power converter circuits using Power Semiconductor Switching Devices	9	Applying
CO4	Develop PLC ladder program for an application	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3		3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Construct various gate triggering methods for SCR		
M1.01	Plot V-I characteristics of SCR	3	Applying
M1.02	Build a single phase control using resistance triggering circuit and 1) plot the waveform across the load and SCR 2) find the maximum firing angle	3	Applying
M1.03	Construct single phase control using RC triggering circuit and 1) plot the waveform across the load and SCR 2) find the maximum firing angle	3	Applying
M1.04	Construct single phase control using UJT triggering circuit and 1) plot the waveform across the load and SCR 2) find the maximum firing angle	3	Applying
CO2	Construct controlled rectifier and power controlling circuits using power semiconductor switching devices		
M2.01	Construct a single-phase control rectifier using SCR with resistive load and 1) find minimum firing angle 2) find maximum firing angle	3	Applying
M2.02	Construct an illumination control circuit using DIAC and TRIAC.	1.5	Applying
M2.03	Construct an SCR-based DC motor speed control circuit.	1.5	Applying

M2.04	Design a time delay relay circuit using UJT and SCR.	3	Applying
	Lab Exam 1	1.5	
CO3	Implement power converter circuits using Power Semiconductor Switching Devices		
M3.01	Construct an oscillator circuit using DIAC and plot the waveform.	3	Applying
M 3.02	Implement an emergency lamp and a battery charger circuit using SCR	3	Applying
M3.03	Implement an inverter circuit using BJT and plot the waveform.	3	Applying
CO4	Develop PLC ladder program for an application		
M4.01	Implement logic gates using PLC.	1	Applying
M4.02	Implement stair case light using PLC.	1.5	Applying
M4.03	Implement water level controller using PLC.	1.5	Applying
M4.04	Implement traffic light controller using PLC.	2	Applying
M4.05	Implement square wave generator using PLC.	1	Applying
M4.06	Implement logic to count pulses from a source and check for the pre-determined value using PLC.	2	Applying
	Open Ended Experiment	3	Applying
	Lab Exam 2	1.5	

**** - Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 3-5. There should be no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.)

Example:

Home Security System – Burglar's

AlarmAutomatic Door

Automatic Vehicle Collision Prevention

SystemChopper circuit using SCR

Text /Reference:

T/R	Book Title/Author
T1	Industrial Electronics and Control - S K Bhattacharya, S Chatterjee.
T2	Programmable logic controllers - Frank D Petruzella
R1	Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - Delmar

Online Resources:

Sl.No	Website Link
1	http://fac.ksu.edu.sa/sites/default/files/lab-manual_v3.pdf
2	http://www.eee.griet.ac.in/wp-content/uploads/2014/12/PLC-Lab-Manual.pdf
3	http://onlinemas.weebly.com/uploads/6/3/9/4/6394050/lab_manual_shafeeq.pdf
4	http://gnindia.dronacharya.info/EEE/Downloads/Labmanuals/Power-Electronics-Lab-Manual.pdf
5	https://www.watelectronics.com/how-to-program-the-programmable-logic-controllers/